S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Practical 08

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

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CODE :

```
GNU nano 6.7                         lru.c                         Modified
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12;
    int frames[3];
    int time[3];
    int i, j, k, pos, min;
    int hit = 0, fault = 0;
    int counter = 0;

    for(i=0;i<3;i++){
        frames[i] = -1;
    }

    for(i=0;i<n;i++){
        int found = 0;

        for(j=0;j<3;j++){
            if(frames[j] == pages[i]){
                hit++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

        if(found == 0){
            fault++;

            pos = 0;
            min = time[0];

            for(j=1;j<3;j++){
                if(time[j] < min){
                    min = time[j];
                    pos = j;
                }
            }

            frames[pos] = pages[i];
            counter++;
            time[pos] = counter;
        }
    }
                                [ Cancelled ]
^G Help      ^O Write Out  ^W Where Is  ^K Cut    ^T Execute  ^C Location   M-U Undo  M-A Set Mark  M-] To Bracket  M-Q Previous  ◂ Back
^X Exit      ^R Read File  ^\ Replace   ^U Paste  ^J Justify  ^/ Go To Line M-E Redo  M-6 Copy      ^Q Where Was    M-W Next      ▸ Forward
```

OUTPUT:-

```
Student@DESKTOP-JJ8CFNB MINGW64 ~
$ nano lru.c

Student@DESKTOP-JJ8CFNB MINGW64 ~
$ gcc lru.c -o lru

Student@DESKTOP-JJ8CFNB MINGW64 ~
$ ./lru
Total Page Hits = 0
Total Page Faults = 12
Page Hit Percentage = 0.00%
Page Fault Percentage = 100.00%

Student@DESKTOP-JJ8CFNB MINGW64 ~
$ |
```